

# THE GATE FRAMEWORK – MASTER GATE FILE (Standalone Edition)

**\*\*A proof-carrying discipline for rigorous reasoning, and a hardening layer for AI-assisted mathematics.\*\***

**\*\*Version:\*\*** Master v1.0 (standalone) · **\*\*Date:\*\*** 2026-07-17 · **\*\*Status:\*\*** current unified set – 13 gates + 1 schema field.

> **\*\*Standalone notice.\*\*** This document is self-contained. A reader with no prior context – human or AI model – can adopt and apply the framework from this file alone. Domain-specific examples are drawn from a Gaussian-random-field proof program; a glossary (§10) defines every specialist term so the examples are legible without that background. Nothing here requires an external file. Another model may treat this as a complete specification and apply it immediately.

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## 0. What this is, and how to use it

### 0.1 The problem it solves

AI-assisted reasoning (and human reasoning under time pressure) fails in a recognizable profile: citations that do not exist or do not say what they are claimed to say; silent conflation of different *kinds* of claim; elegant arguments that are load-bearing and wrong; numbers presented as measurements that were never measured. These failures are individually plausible and collectively fatal. The gate framework is a set of **\*\*mechanically checkable rules\*\*** that catch this failure profile *before* a claim is trusted or promoted.

### 0.2 The unit of operation

The framework operates on **\*\*claims\*\*** – the atomic assertions (nodes, lemmas, positions, results, estimates) in a reasoning artifact. Every claim is written as a structured **\*\*proof object\*\*** (§2) carrying explicit metadata. The gates (§4) are predicates over proof objects and the dependency graph they form.

### 0.3 The core discipline, in one sentence

> **\*\*Name the restriction, gate the drop.\*\***

Every claim is established *under restrictions* – a fixed regime, a specific measure, a single side, an independence assumption, a finite parameter range, a charted region. A claim fails when a restriction under which it was established is silently dropped when it is reused. Each gate guards one restriction axis and forbids its silent drop. This is the organizing principle of the entire set (§1), and it is the rule for extending the set (§6).

### ### 0.4 How to apply it

1. Write every externally visible claim as a proof object with all fields populated (§2).
2. Assign each claim an **epistemic grade** and record its dependencies (§3.1).
3. Run **all** gates over the claim and its dependency cone (§4). A gate returns PASS, FAIL, or PROVISIONAL.
4. A claim may be **promoted** (used as a dependency, or published at its stated grade) **only if every gate passes.** A FAIL blocks promotion until repaired; a PROVISIONAL caps the claim's grade (§4.12-4.13).
5. Build thresholds, bands, and estimators only as the gate-design standard permits (§5).
6. Preserve every failure (§7). Corrections supersede; they never overwrite.
7. To add a gate, use the retrodiction protocol (§6) – never a constructed or forward-looking score.

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### ## 1. The meta-rule: restriction-drop

The gates are not independent inventions. They are instances of a single shape:

> **A property established under a restriction must not be used after the restriction is silently dropped. Either carry the restriction, or supply an explicit bridge that licenses its removal.**

Each gate guards one **restriction axis**:

|                  |                                                                   |
|------------------|-------------------------------------------------------------------|
| gate             | restriction axis it guards                                        |
| ---              | ---                                                               |
| POLARITY         | sidedness (upper / lower / two-sided)                             |
| MEASURE          | base measure / conditioning law                                   |
| MARK             | mark / type carried by the estimate                               |
| MODEL            | the exact probabilistic model (e.g. covariance)                   |
| RUNG             | the scale a measured value lives at                               |
| COMPOSITION      | statistical independence (or the witness that licenses combining) |
| DOMAIN           | the parameter range of validity                                   |
| ENDPOINT         | verification at the boundary of a declared range                  |
| COVERAGE         | the region actually charted by a detector                         |
| PRECEDENCE       | live-vs-dead (killed / superseded) status                         |
| CORE CLOSURE     | finished-vs-open status                                           |
| COMMON-MODE      | independence of the <i>*checking instruments*</i> error channels  |
| HEURISTIC-BRIDGE | rigor mode (heuristic/analogy vs. derivation)                     |

**Why this framing matters.** It makes the set memorable, teachable, and extensible. A new gate is justified exactly when the failure record exhibits a *\*restriction axis not yet guarded\** (§6). It also makes the set diagnosable: to

audit a claim, enumerate the restrictions its derivation actually relied on, and check each drop.

**Honesty on the framing.** The fit is clean for most gates and loose for a few (PRECEDENCE, CORE CLOSURE, and ENDPOINT read as restriction-drop only by stretch). Treat §1 as an **organizing lens**, not a theorem – a structural analogy across the gates, offered as such.

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## ## 2. The proof object (claim schema)

Every externally visible claim carries these fields. A missing field is itself a gate failure (a claim that cannot state its own metadata cannot be trusted).

| field                                  | meaning                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>atomic statement</b>                | the single assertion, stated without hedging or bundling                                                                                                                                                                                                                |
| <b>dependency IDs</b>                  | the claims this one rests on (edges into the dependency graph)                                                                                                                                                                                                          |
| <b>evidence IDs</b>                    | the computations, proofs, or measurements that support it                                                                                                                                                                                                               |
| <b>model tag</b>                       | the exact probabilistic/mathematical model assumed (§3.3)                                                                                                                                                                                                               |
| <b>measure tag</b>                     | the base measure / conditioning law (§3.2)                                                                                                                                                                                                                              |
| <b>parameter domain</b>                | the range of every free parameter over which the claim holds                                                                                                                                                                                                            |
| <b>required marks / supplied marks</b> | the type/mark variables the claim <i>needs</i> vs. the ones its evidence <i>carries</i> (§3.4)                                                                                                                                                                          |
| <b>quantifier</b>                      | $\forall$ / $\exists$ / "for typical" – stated, not implied                                                                                                                                                                                                             |
| <b>direction</b>                       | lower bound, upper bound, or equality/two-sided                                                                                                                                                                                                                         |
| <b>epistemic grade</b>                 | position on the warrant ladder (§3.1)                                                                                                                                                                                                                                   |
| <b>uncertainty side</b>                | for a numerical band: one-sided (+ or -) or symmetric, with justification                                                                                                                                                                                               |
| <b>scale / rung tag</b>                | the scale a measured value is expressed in (§3.5)                                                                                                                                                                                                                       |
| <b>composition witness</b>             | if the claim combines sub-claims, the named license for combining (§3.6)                                                                                                                                                                                                |
| <b>source-precedence ID</b>            | which version/supersession lineage this claim belongs to                                                                                                                                                                                                                |
| <b>content hash</b>                    | a hash of the claim's exact text/data, for integrity and citation                                                                                                                                                                                                       |
| <b>supersession status</b>             | live / superseded / killed / retired                                                                                                                                                                                                                                    |
| <b>regime-of-validity</b>              | $*(field\ F-1, added\ 2026-07-17)*$   the tuple (base measure, spatial region, scale, parameter box, conditioning depth) under which the claim/mechanism/constant was <i>established</i> – so transfer gates have an explicit deployment target to check against (§3.7) |

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## ## 3. Controlled lexicons

Gates are only as sharp as the vocabularies they check against. These lexicons are mandatory and closed (extend only with an explicit, consistently applied definition).

### ### 3.1 Epistemic grade ladder (warrant axis)

From strongest to weakest warrant:

1. **Primitive** – a declared assumption (recorded in an assumption ledger). Not proved; explicitly assumed.
2. **Proven** – proved within the artifact (referee-grade), or a self-contained theorem package.
3. **Proven-Modulo** – proved *given* named conditional dependencies; the conditions are explicit and finite.
4. **Derived** – calculation or bound executed, not yet packaged to referee grade.
5. **Literature-Supported** – supported by cited external work; may require adaptation.
6. **Plausible** – coherent and mechanism-supported, but not proof-complete.
7. **Conjecture** – believed, not proved.
8. **Open** – explicitly unresolved.
9. **Killed** – false, invalid, superseded, or unusable.

**Constitutional dependency rule (binding).** A claim may depend **only** on objects graded Primitive, Proven, Proven-Modulo, or Literature-Supported. It may **never silently** depend on Plausible, Conjecture, Open, or Killed. If it depends on an Open or Conjecture object, the claim **must** itself be graded Proven-Modulo, making the conditional dependence explicit. A Killed object may never re-enter the dependency graph unless repaired and relabeled. This single rule prevents hidden circularity and silent dependence on dead mathematics.

**Note on axes.** This ladder measures *warrant* (how well-established). It is distinct from *reasoning type* (empirical / theoretical / analogical / normative). Where the two interact – an analogical step becoming load-bearing – see gate 13 (HEURISTIC-BRIDGE).

### ### 3.2 Measure / law-type lexicon

The conditioning law under which a probabilistic quantity is computed. These may not be silently interchanged (gate MEASURE):

- **Unconditioned** – the plain law.
- **Gaussian-pinned** – conditioned on field values (and/or derivatives) at specified points.
- **Typed-pinned** – conditioned on critical points of a specified index/type.
- **Pair-Palm** – the Palm law of a *pair* point process (conditioned on the existence of a configured pair at specified locations).

\*(A general adopter substitutes the conditioning laws native to their domain; the rule – do not interchange without a bridge – is invariant.)\*

### ### 3.3 Model lexicon

The exact model a claim assumes must be named (gate MODEL): the specific kernel/covariance, distribution family, or generative model – or, if a claim is transported from one model to another, an explicit **transfer certificate** bounding the error of the transport, valid within a declared scope.

### ### 3.4 Mark / type lexicon

A **mark** is auxiliary data attached to a point/event (e.g. height/value, critical-point index/type). An estimate is **marked** if it controls the marked quantity and **unmarked** if it controls only the underlying (e.g. spatial) quantity. An unmarked estimate cannot discharge a marked obligation (gate MARK).

### ### 3.5 Scale / rung lexicon

Every measured value carries the scale it is expressed in (e.g. absolute units, or units of a small parameter such as  $r$ ,  $r^2$ ,  $\ell$ ). A measured number with no scale tag is uninterpretable and may not be compared or composed (gate RUNG).

### ### 3.6 Composition-witness lexicon

When a claim combines sub-claims (a product of probabilities, a union bound, an assembly), the license must be one of these **named** witnesses (gate COMPOSITION):

- **Independence** – the events are independent.
- **Conditional independence** – independent given a stated conditioning.
- **Markov property** – a stated Markov structure.
- **Negative dependence / negative association** – a stated inequality direction.
- **Comparison theorem** – a named inequality (e.g. Slepian, FKG, Gaussian comparison).
- **Joint certificate** – a direct bound on the joint object.
- **Exact algebra** – the combination is an identity, not an approximation.

A product/union with **no** witness is forbidden.

### ### 3.7 Regime-of-validity (field F-1)

The tuple recording where a claim/constant/mechanism was **established**: base measure, spatial region/scale, parameter box, conditioning depth. Its purpose is to give the transfer gates (MEASURE, DOMAIN, MODEL, MARK) an explicit target: a deployment site is checked against the establishment regime, and a mismatch is flagged. This is a **schema field**, not a gate – the minimal fix for the most frequent transfer-error shape without blurring the four sharp transfer gates into one vague one.

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## ## 4. The gates (complete, uncompressed)

Each gate is stated as: **\*\*Guards\*\*** (restriction axis) · **\*\*Rule\*\*** · **\*\*PASS / FAIL condition\*\*** (mechanically checkable) · **\*\*Canonical failure prevented\*\*** · **\*\*Notes\*\***. Gates 1-11 are the established set; gates 12-13 are added 2026-07-17 with the retrodiction evidence in the appendix.

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### ### Gate 1 – DOMAIN

- **\*\*Guards:\*\*** the parameter range of validity.
- **\*\*Rule:\*\*** a dependency must cover the complete parameter domain of the claim that invokes it.
- **\*\*PASS / FAIL:\*\*** PASS iff, for every free parameter, the dependency's domain  $\supseteq$  the parent's domain. FAIL if the parent is asserted on a range the dependency does not cover.
- **\*\*Canonical failure prevented:\*\*** a lemma proved on a sub-interval used to support a claim asserted on a wider interval; a bound established for one parameter box invoked outside it.
- **\*\*Notes:\*\*** domain containment is checked with an explicit tolerance; do not rely on "obviously covers."

### ### Gate 2 – ENDPOINT

- **\*\*Guards:\*\*** verification at the boundary of a declared range.
- **\*\*Rule:\*\*** every registered endpoint of a claimed finite-range bound must satisfy that bound.
- **\*\*PASS / FAIL:\*\*** PASS iff evaluating the claim at each registered endpoint satisfies the stated inequality. FAIL if any endpoint violates it.
- **\*\*Canonical failure prevented:\*\*** asserting a ceiling (e.g. " $\leq 0.97$  on this range") whose range secretly contains a point where the value is higher (e.g. 0.985) – the bound is false at the boundary it claims to cover.
- **\*\*Notes:\*\*** endpoints are the cheapest falsifiers of a range claim; check them first.

### ### Gate 3 – POLARITY

- **\*\*Guards:\*\*** sidedness.
- **\*\*Rule:\*\*** a lower-bound claim may consume only lower one-sided inputs; an upper-bound claim only upper one-sided inputs; and any uncertainty band attached to a one-sided claim must itself be one-sided in the matching direction.
- **\*\*PASS / FAIL:\*\*** PASS iff the direction of every input matches the direction of the claim, and every band on a one-sided claim is one-sided (+ for upper, - for lower) unless the inputs are proved to be already one-sided limits. FAIL on a symmetric band attached to a one-sided claim, or a lower input feeding an upper claim (or vice versa).
- **\*\*Canonical failure prevented:\*\*** attaching a symmetric  $\pm\epsilon$  band to an upper theorem (an upper claim consumes only the + side); requiring a two-sided bound where only one side is needed.
- **\*\*Notes:\*\*** " $\pm$ " on a one-sided quantity is the most common tell.

### ### Gate 4 – MEASURE

- **\*\*Guards:\*\*** base measure / conditioning law.
- **\*\*Rule:\*\*** unconditioned, Gaussian-pinned, typed-pinned, and pair-Palm claims (or the domain's analogous conditioning laws) may not be silently interchanged.
- **\*\*PASS / FAIL:\*\*** PASS iff the measure tag of each input matches the measure the claim requires, or a stated change-of-measure bridges them. FAIL on any silent swap of conditioning law.
- **\*\*Canonical failure prevented:\*\*** using a coefficient computed under one conditioning (e.g. a pinned weight) where the claim requires another (e.g. the Palm weight) – a real recorded instance had these differ by nearly a factor of two.
- **\*\*Notes:\*\*** the conditioning law is part of *\*what the number means\**; a bare number hides it.

### ### Gate 5 – MARK

- **\*\*Guards:\*\*** the mark/type carried by the estimate.
- **\*\*Rule:\*\*** a spatial (unmarked) estimate does not prove a height-marked or type-marked estimate unless the mark variables appear in the dependency cone.
- **\*\*PASS / FAIL:\*\*** PASS iff every required mark of the claim is supplied by its evidence. FAIL if a marked obligation is discharged by unmarked support.
- **\*\*Canonical failure prevented:\*\*** citing spatial repulsion between features (unmarked) to assert a bound that requires a *\*height-window\** factor (marked); the unmarked fact does not, by itself, restore the marked factor.
- **\*\*Notes:\*\*** "incomplete source-to-estimand transfer" – the source proves less than the claim needs – is the general disease; MARK is its most common form.

### ### Gate 6 – COMPOSITION

- **\*\*Guards:\*\*** statistical independence (or the license to combine).
- **\*\*Rule:\*\*** a product of probabilities, a union bound, or any assembly of sub-claims requires a named composition witness (§3.6).
- **\*\*PASS / FAIL:\*\*** PASS iff every combination carries a witness from the closed list. FAIL on any witness-free product/union/assembly.
- **\*\*Canonical failure prevented:\*\*** modeling a multi-checkpoint corridor probability as the product of per-checkpoint probabilities with no dependence structure (a recorded instance was off by up to ~3.35×); treating a maximum/determinant as independent of its conditioning.
- **\*\*Notes:\*\*** independence is a *\*claim\**, not a default; if it cannot be named, it must be bounded.

### ### Gate 7 – RUNG

- **\*\*Guards:\*\*** the scale of a measured value.
- **\*\*Rule:\*\*** every measured value carries an explicit scale/parameter tag.
- **\*\*PASS / FAIL:\*\*** PASS iff each measured number states the scale it is expressed in. FAIL if a measured value is treated as scale-free or compared across mismatched scales.
- **\*\*Canonical failure prevented:\*\*** promoting a measured constant to a "truth

constant" with no scale, then composing it with quantities at a different scale.

- **\*\*Notes:\*\*** pairs with F-1 (§3.7): the rung is one component of the full regime-of-validity.

### ### Gate 8 – PRECEDENCE

- **\*\*Guards:\*\*** live-vs-dead status.
- **\*\*Rule:\*\*** a killed or superseded claim may not be reachable from a live root in the dependency graph.
- **\*\*PASS / FAIL:\*\*** PASS iff no path from a live theorem reaches a node marked killed/superseded/retired. FAIL on any such reachable path.
- **\*\*Canonical failure prevented:\*\*** continuing to cite a retired estimand or an intermediate constant that a later cycle superseded.
- **\*\*Notes:\*\*** enforce by graph reachability on the supersession status field; supersede-never-overwrite (§7) keeps the dead nodes present *and* marked.

### ### Gate 9 – CORE CLOSURE

- **\*\*Guards:\*\*** finished-vs-open status.
- **\*\*Rule:\*\*** a declared-finished core may not depend on an open extension node.
- **\*\*PASS / FAIL:\*\*** PASS iff no claim graded as part of a closed core has a dependency graded Open (without the constitutional Proven-Modulo relabel). FAIL otherwise.
- **\*\*Canonical failure prevented:\*\*** declaring a result "closed" while it silently rests on an unresolved lemma.
- **\*\*Notes:\*\*** the mirror of the constitutional rule (§3.1) at the level of whole cores.

### ### Gate 10 – MODEL

- **\*\*Guards:\*\*** the exact probabilistic model.
- **\*\*Rule:\*\*** every model-dependent (e.g. covariance-dependent) claim identifies its exact model, or supplies an explicit transfer certificate valid within a declared scope.
- **\*\*PASS / FAIL:\*\*** PASS iff the model tag is present and matches the claim's evidence, or a transfer certificate with a stated scope bridges them. FAIL on an unnamed model, or a transfer used outside its certified scope.
- **\*\*Canonical failure prevented:\*\*** a covariance computation that does not name its kernel; a model-transfer certified only for a bounded set of orders/dimensions/distances but invoked outside that box (the fix: partition the domain and declare any larger use).
- **\*\*Notes:\*\*** "which model?" must have a single, checkable answer.

### ### Gate 11 – COVERAGE

- **\*\*Guards:\*\*** the region actually charted by a detector.
- **\*\*Rule:\*\*** no aggregate/exponential-rank argument may suppress wrong outputs in a region that activates **\*\*no\*\*** registered detector (chart). Strength elsewhere does not cover an un-instrumented region.
- **\*\*PASS / FAIL:\*\*** PASS iff every region where the claim could fail is covered by at least one detector. FAIL if a strong global argument is used to imply control

over a region with no detector.

- **\*\*Canonical failure prevented:\*\*** a suppression bound that is exponentially strong on the charted set being read as control over a qualitatively different, un-charted failure region (a coverage failure with no exponential suppression).
- **\*\*Notes:\*\*** the sibling of gate 12 – COVERAGE asks **"is the region charted at all?"**; COMMON-MODE asks **"are the charts that agree actually independent?"**

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### Gate 12 – COMMON-MODE \*(added 2026-07-17)\*

- **\*\*Guards:\*\*** independence of the **\*checking instruments\*** error channels.
- **\*\*Rule:\*\*** a gate PASS that **\*\*rests on agreement between two instruments\*\*** (any regression/agreement check comparing two computations) is valid only if the instruments are shown not to share the error channel under test – via either (a) a second construction with a demonstrably disjoint failure mode, or (b) an external contradiction test against an independent artifact. Absent (a) or (b), the PASS is **\*\*PROVISIONAL\*\*** and the claim it supports may not promote in grade.
- **\*\*PASS / FAIL / PROVISIONAL:\*\*** PASS iff an agreement-based verdict carries an error-independence argument (a) or an external cross-check (b). PROVISIONAL if the verdict rests on agreement alone. Single-instrument exact (arbitrary-precision) checks are **\*\*exempt\*\*** – they assert no agreement to certify.
- **\*\*Canonical failure prevented:\*\*** two instruments that share a construction step agree, and the agreement is mistaken for certification, when both carry the same defect – e.g. two computations that both include a spurious point inside a search window, or both drop the same coordinate offset and therefore "agree at the wrong station." Recorded instances were caught in reality only by luck or by an external cross-check; this gate makes that cross-check a condition of PASS rather than a fortunate catch.
- **\*\*Notes:\*\*** all of gates 1-11 constrain the structure of a **\*claim\***; this is the first gate that constrains **\*verification methodology\***. Scope it strictly to agreement-based verdicts to avoid a blanket "reimplement everything" tax. The dual use is also valid and encouraged: when two independent constructions produce the **\*same wrong\*** result, that shared error convicts the shared instrument, not the object.

### Gate 13 – HEURISTIC-BRIDGE \*(added 2026-07-17)\*

- **\*\*Guards:\*\*** rigor mode (heuristic/analogy vs. derivation).
- **\*\*Rule:\*\*** any step whose warrant is analogy, structural similarity, or an unexplained symbolic coincidence, and which becomes **\*\*load-bearing for a quantitative or asymptotic claim\*\***, must carry an explicit **\*\*bridge\*\*** (a derivation of the transition) or an explicit **\*\*grade downgrade\*\*** of the dependent claim. An un-bridged heuristic-quantitative transition may not promote past **\*\*Plausible\*\***.
- **\*\*PASS / FAIL / PROVISIONAL:\*\*** PASS iff each analogical/heuristic step feeding a quantitative claim is either bridged or the dependent claim is graded  $\leq$  Plausible. FAIL if an analogical step silently supports a claim graded Derived or higher without a bridge.

- **Canonical failure prevented:** an argument by analogy to a different theory, treated as a derivation and made to carry a quantitative prediction (later falsified by measurement); a symbolic derivation in which required quantities appear "as if by magic" with no accounting (the fix supplies the missing derivation).

- **Notes:** this is the guard on the axis where *warrant* (§3.1) and *reasoning type* interact. It bites at the generative frontier – precisely where the most expensive errors originate. Its cost (a tag plus a bridge-or-downgrade obligation) is low.

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## ## 5. Gate-design standard (how to build thresholds, bands, and estimators)

Gates and their thresholds are themselves artifacts that can be wrong. These rules (registry-grade) govern their construction:

1. **Every threshold must be DERIVED** – from a physical/structural scale, or from the estimator's sampling distribution. **Never** compare single-instance statistics against arbitrary percent bands, and never use an arbitrary absolute tolerance.
2. **Separate exactness from deployment.** Deterministic claims get arbitrary-precision (exact) gates, kept separate from floating-point deployment gates.
3. **Diagnose the instrument before the physics.** When a gate FAILS, first suspect the variable definition and the estimator; interrogate those before concluding a substantive/physical failure. (A recorded case: a first diagnosis of "aliasing" was wrong; the true cause was an under-resolved grid.)
4. **Limit-bands from remainder structure only.** A band on a limiting value must come from the exact remainder/error structure, or be explicitly labeled a *forecast* – never from naive extrapolation of finite-parameter data with competing correction terms.
5. **Independent numerical check before comparison.** A closed-form constant gets an independent numerical evaluation before it is used in any comparison (guards against hand-arithmetic and transcription error).

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## ## 6. Adding a new gate – the retrodiction protocol

New gates are earned against a **frozen failure record**, never justified by a constructed or forward-looking score. (A forward-looking "improvement percentage" over cases selected to flatter the gate is not evidence; it is arithmetic that follows from the selection.)

**Protocol.** Freeze a record *F* of past failures (a kill registry / error ledger). A candidate rule *R* qualifies only if:

1. ***R* fires on  $\geq 2$  independent recorded entries** in *F* – it would have flagged

them before their actual catch mechanism did.

2. **R** is silent on most of **F** – a rule that flags everything is uninformative.
3. **R** is not already discharged by an existing gate or by the gate-design standard.

If any condition fails, **reject R**. Most candidates die on condition 3 – and rejecting them is the point. The goal is not to reach a target count of gates; the number of gates is an **output** of what the failure record demands, not an input.

**Binding limitation (survivorship).** Retrodiction tests only *caught* failures. A gate's value against failures that were never detected is unmeasurable this way. Retrodiction can show a rule would have helped on known misses; it cannot prove a rule's full value, nor that the current gap list is complete. State this limitation whenever the framework is presented as validated.

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## ## 7. Operating disciplines

The gates presuppose a workflow that preserves state and failures:

- **Freeze-before-execute.** Commit gate specifications and predicted outcomes *before* running the instrument. A prediction written after seeing the result is not a test.
- **Supersede-never-overwrite.** A correction creates a new versioned artifact; the superseded artifact is retained and marked. This is what lets PRECEDENCE (gate 8) enforce reachability on dead nodes.
- **Preserve every failure.** Every kill and every failed-as-written gate is logged permanently in an error ledger (a "kill registry" / E-ledger). Failures are structure, not embarrassment: they are the test set for \$6.
- **Ledgers written after adjudication.** Record gate outcomes in a step *separate from and subsequent to* running the gates. A ledger written in the same act as the computation can claim a result before it is inspected.
- **Adversarial reimplementations.** Verify by independent instruments – and, per gate 12, with a demonstrably disjoint error channel where a verdict rests on agreement.
- **Separation of duties.** Distinct roles across generation, verification, and adjudication; dual-track (external citation/source verification in parallel with internal consistency checking).
- **Maximize the weakest link.** A claim's grade is bounded by its weakest dependency; a chain is only as exact as its least-exact link. Design for the weakest link, not the average.

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## ## 8. Minimal worked example

**\*\*Claim (as written):\*\*** "The probability that a configured feature fails to pair with its nearest partner is at most  $C \cdot r^3$ , for  $r \in (0, r_0]$  and parameter  $b \in B$ ."

**\*\*Proof object (populated):\*\***

- atomic statement: as above.
- direction: upper bound. · quantifier:  $\forall r \in (0, r_0], \forall b \in B$ .
- measure tag: pair-Palm. · model tag: [named kernel]. · parameter domain:  $r \in (0, r_0], b \in B$ .
- required marks: height  $b$  (marked). · supplied marks: must include  $b$ .
- epistemic grade: Proven-Modulo. · dependencies: {inner-region bound, far-region bound, boundary bound}.
- uncertainty side: upper (+) only. · scale/rung: coefficient in units of  $r^3$ .
- composition witness: how the region bounds combine. · regime-of-validity (F-1): (pair-Palm, near-diagonal region,  $r$ -scale, box  $r \leq r_0$  &  $b \in B$ , full pair conditioning).

**\*\*Running the gates (illustrative):\*\***

- **\*\*POLARITY\*\*** – upper claim, upper inputs, one-sided (+) band  $\rightarrow$  candidate PASS.
- **\*\*MEASURE\*\*** – every regional input must be pair-Palm (not merely pinned). If one input is a \*pinned\* coefficient, MEASURE **\*\*FAILS\*\*** until a change-of-measure bridge is supplied. \*(This is the schema catching a real, common error.)\*
- **\*\*MARK\*\*** – the height  $b$  must be carried by the evidence; if a region bound is spatial-only, MARK **\*\*FAILS\*\***.
- **\*\*COMPOSITION\*\*** – the three regional bounds are combined; the combination needs a witness (e.g. a union bound with disjoint events, or a comparison theorem). No witness  $\rightarrow$  **\*\*FAIL\*\***.
- **\*\*DOMAIN\*\*** – each regional bound must cover  $r \in (0, r_0]$  and  $b \in B$ ; a bound proved only for  $r \leq r_0/2 \rightarrow$  **\*\*FAIL\*\***.
- **\*\*COMMON-MODE\*\*** – if the coefficient  $C$  was certified by two numerical instruments agreeing, and both share the same sampling construction, the PASS is **\*\*PROVISIONAL\*\***;  $C$  may not be promoted until an error-independent check or external cross-check is added.
- **\*\*HEURISTIC-BRIDGE\*\*** – if any regional bound rests on an analogy to another model rather than a derivation, the claim may not be graded above **\*\*Plausible\*\*** until the analogy is bridged.

**\*\*Verdict.\*\*** The claim promotes to Proven-Modulo only when every gate passes; otherwise its grade is capped or promotion is blocked at the named gate, and the failure is logged (§7).

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**## 9. The gates at a glance (checklist form)**

For rapid application, a claim is gated by answering each:

1. **\*\*DOMAIN\*\*** – do all dependencies cover the full parameter range asserted?

2. **\*\*ENDPOINT\*\*** – does the bound hold at every boundary of its declared range?
3. **\*\*POLARITY\*\*** – do input directions and band sides match a one-sided claim?
4. **\*\*MEASURE\*\*** – is every input under the required conditioning law (no silent swap)?
5. **\*\*MARK\*\*** – are all required marks/types supplied by the evidence?
6. **\*\*COMPOSITION\*\*** – does every product/union/assembly carry a named witness?
7. **\*\*RUNG\*\*** – does every measured value carry its scale?
8. **\*\*PRECEDENCE\*\*** – is any killed/superseded node reachable from this claim?
9. **\*\*CORE CLOSURE\*\*** – does a "finished" claim secretly depend on an open node?
10. **\*\*MODEL\*\*** – is the exact model named (or a scoped transfer certificate supplied)?
11. **\*\*COVERAGE\*\*** – is every possible-failure region covered by a detector?
12. **\*\*COMMON-MODE\*\*** – if a PASS rests on instrument agreement, are the instruments' errors independent?
13. **\*\*HEURISTIC-BRIDGE\*\*** – is any load-bearing analogy bridged, or the claim graded  $\leq$  Plausible?

Plus the schema field **\*\*F-1\*\*** (regime-of-validity) on every constant/mechanism, and the **\*\*constitutional dependency rule\*\*** (§3.1) on every edge.

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## ## 10. Glossary (for readers outside the source domain)

- **\*\*Palm distribution / pinning\*\*** – a conditioned law: the distribution of a random field *\*given\** that specified events occur (e.g. given field values at points, or given a critical point of a given type at a location). Different conditionings give genuinely different laws.
- **\*\*Pair-Palm\*\*** – the Palm law conditioned on the existence of a *\*pair\** of configured features at specified places.
- **\*\*Kac-Rice\*\*** – a formula giving the expected number/intensity of level crossings or critical points of a random field; the standard tool for such counts.
- **\*\*Mark / marked point process\*\*** – points carrying auxiliary data (a "mark"), e.g. the height/value of the field or the type/index of a critical point. Controlling the unmarked points is weaker than controlling the marked ones.
- **\*\*Model / covariance / kernel\*\*** – the exact stochastic model; for a Gaussian field, its covariance function (kernel) determines everything. "Which model" must be named.
- **\*\*Arbitrary-precision (exact) vs. floating-point\*\*** – exact symbolic/rational or very-high-precision computation, as opposed to ordinary floating-point; the framework keeps their gates separate.
- **\*\*Freeze\*\*** – committing a specification/prediction to an immutable record before executing the computation that tests it.
- **\*\*Kill registry / E-ledger\*\*** – the permanent, append-only record of failed claims, preserved as the test set for validating gates.
- **\*\*Retrodiction\*\*** – testing a proposed rule against a *\*frozen\** record of past failures (can it return "no"?), as opposed to a constructed or forward-looking

score.

- **\*\*Supersede-never-overwrite\*\*** – corrections create new versions; superseded artifacts are retained and marked, never deleted.
- **\*\*Restriction-drop\*\*** – the framework's core failure shape: using a result after silently dropping a restriction under which it was established.

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## ## Appendix A – Provenance and change log

- **\*\*Gates 1-11\*\*** (DOMAIN, ENDPOINT, POLARITY, MEASURE, MARK, COMPOSITION, RUNG, PRECEDENCE, CORE CLOSURE, MODEL, COVERAGE): established set, developed through successive audit cycles; the semantic gates were consolidated and machine-checked in a verifier prototype; COVERAGE was the most recent of these, from a late audit finding an un-charted failure region.
- **\*\*Gate 12 (COMMON-MODE)\*\***: added 2026-07-17. Justified by retrodiction against the frozen kill record – it fires on recorded instances where two instruments sharing a construction step agreed while both carried the same defect (a spurious point inside a search window; a shared dropped coordinate offset), and is silent on the rest of the record. The principle had been *\*stated in prose\** and *\*used correctly\** in earlier cycles but never codified – the same "noticed, then promoted to a standing rule" path that produced the gate-design standard (§5).
- **\*\*Gate 13 (HEURISTIC-BRIDGE)\*\***: added 2026-07-17. Justified by retrodiction – it fires on recorded kills where an analogy to another theory, or an unexplained symbolic step, was made load-bearing for a quantitative claim, and is silent on dependence/definition/order errors.
- **\*\*Field F-1 (regime-of-validity)\*\***: added 2026-07-17 as a schema field (not a gate) – the minimal fix for the most frequent transfer-error shape (a mechanism validated in one regime deployed in another), giving the transfer gates an explicit deployment target without collapsing them into one vague gate.

**\*\*Count note.\*\*** The set stands at 13 gates + 1 schema field not because a count was targeted, but because the frozen failure record yields exactly two previously-un-gated restriction axes (verification-instrument independence; heuristic-vs-derivation rigor mode) and one un-tagged regime field. Future additions must clear the retrodiction protocol (§6); the count is an output of the failure record, not a goal.

## ## Appendix B – Adoption notes for another model

- Treat this file as a complete specification. To apply it: represent each claim as a §2 proof object; assign a §3.1 grade; run the §9 checklist / §4 gates over the claim and its dependency cone; promote only on all-pass; cap grade or block on any FAIL/PROVISIONAL; log failures per §7.
- Do **\*\*not\*\*** weaken the quantitative gates into qualitative metaphors when applying the framework to a soft domain. The framework's value is that its checks are mechanical; a gate you cannot check mechanically is not a gate.

- When reporting the framework's efficacy, report it as a **\*\*qualitative finding tied to specific caught cases\*\***, with the survivorship limitation stated (§6). Do not report a numerical "improvement" over cases selected to be favorable – that is the exact failure mode this framework exists to catch.

\*End of Master Gate File – 2026-07-17.\*